Ideation Phase
Define the Problem Statements

Date	04 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	2 Marks

Problem Statements — Power Consumption Dataset:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A load-despatch planner at a State Electricity Board, responsible for daily demand forecasting	predict next-day and seasonal power demand accurately for the state	cannot spot clear seasonal and day-wise consumption trends from the raw state-wise usage records	the data is scattered across hundreds of state-date rows with no visual trend or seasonal summary	anxious about under-provisioning during peak periods or over-provisioning and wasting resources
PS-2	A regional grid operator managing load balance across the NR, WR, SR, ER and NER regions	identify which region is approaching peak demand so power can be rerouted before overload	cannot easily compare live usage trends between the five regions side by side	usage figures are recorded per state per day and are not aggregated or benchmarked at the regional level	stressed and constantly on edge about the risk of grid failure or blackouts
PS-3	A renewable-energy policy analyst at a central ministry, deciding where to expand solar and wind capacity	find which states show the fastest-growing electricity consumption over time	cannot see long-term growth patterns or compare states against each other at a glance	the dataset only contains raw daily figures with no year-over-year or state-comparison analysis	uncertain and hesitant about committing investment budgets to the right locations
PS-4	A large industrial consumer's energy manager tracking factory power costs across multiple plant locations	plan monthly energy budgets and negotiate better tariffs by anticipating consumption swings	cannot predict when usage will spike due to seasonal or regional demand shifts	there is no simple dashboard that turns historical consumption data into forward-looking insight	frustrated by unexpected cost overruns that are hard to explain to leadership
PS-5	A data science student building a smart-grid forecasting project as part of a course	build a reliable model that predicts future state-wise power consumption	struggles to extract clean seasonal, regional, and geographic patterns from the raw CSV	the data is unaggregated, mixing states, regions, coordinates, and dates in a single flat file	overwhelmed by the complexity of the data before any modelling can even begin